

# CliniScan

Chest X-Ray Analysis System

## Milestone 2 — Model Training Report

|             |                                        |
|-------------|----------------------------------------|
| Intern Name | Nimisha Sharma                         |
| Project     | B13-CliniScan                          |
| Milestone   | 2 — Model Training                     |
| Date        | 5 March 2026                           |
| Hardware    | AMD Ryzen 3 3250U   8GB RAM   CPU Only |

*This report documents the training of two deep learning models for chest X-ray analysis: a Classification Model using EfficientNet-B0 and a Detection Model using YOLOv8s. Both models were trained entirely on CPU hardware.*

# 1. Project Overview

CliniScan is an AI-powered chest X-ray analysis system that performs two complementary tasks:

| Model          | Task                              | Architecture    | Output                              |
|----------------|-----------------------------------|-----------------|-------------------------------------|
| Classification | Predict which abnormalities exist | EfficientNet-B0 | 14-class multi-label probabilities  |
| Detection      | Locate abnormalities with boxes   | YOLOv8s         | Bounding boxes + class + confidence |

Both models were trained on a subset of 5,000 images from the VinBigData Chest X-ray Abnormalities Detection dataset, preprocessed from DICOM format to 224x224 PNG images.

# 2. Dataset

## 2.1 Source

VinBigData Chest X-ray Abnormalities Detection (Kaggle). Images converted from DICOM to 224x224 PNG during Milestone 1 preprocessing.

## 2.2 Class Distribution

The dataset contains 14 abnormality classes plus a No Finding class:

| Class ID | Class Name         | Type       |
|----------|--------------------|------------|
| 0        | Aortic enlargement | Cardiac    |
| 1        | Atelectasis        | Pulmonary  |
| 2        | Calcification      | Structural |
| 3        | Cardiomegaly       | Cardiac    |
| 4        | Consolidation      | Pulmonary  |
| 5        | ILD                | Pulmonary  |
| 6        | Infiltration       | Pulmonary  |
| 7        | Lung Opacity       | Pulmonary  |
| 8        | Nodule/Mass        | Structural |
| 9        | Other lesion       | Other      |
| 10       | Pleural effusion   | Pleural    |

| Class ID | Class Name          | Type      |
|----------|---------------------|-----------|
| 11       | Pleural thickening  | Pleural   |
| 12       | Pneumothorax        | Pleural   |
| 13       | Pulmonary fibrosis  | Pulmonary |
| 14       | No finding (Normal) | Normal    |

2.3 Dataset Split

| Split      | Images | Percentage | Purpose               |
|------------|--------|------------|-----------------------|
| Train      | 3,500  | 70%        | Model learning        |
| Validation | 750    | 15%        | Hyperparameter tuning |
| Test       | 750    | 15%        | Final evaluation      |
| Total      | 5,000  | 100%       |                       |

### 3. Classification Model

#### 3.1 Architecture

| Component     | Detail                          | Reason                                        |
|---------------|---------------------------------|-----------------------------------------------|
| Base Model    | EfficientNet-B0                 | 4x faster than ResNet50 on CPU, same accuracy |
| Pretrained On | ImageNet (DEFAULT weights)      | Transfer learning for better performance      |
| Final Layer   | Linear(1280 → 14)               | Adapted for 14-class multi-label output       |
| Activation    | Sigmoid (via BCEWithLogitsLoss) | Independent probability per class             |
| Loss Function | BCEWithLogitsLoss               | Multi-label classification standard           |
| Optimizer     | Adam (lr=0.001)                 | Adaptive learning rate                        |
| Scheduler     | ReduceLROnPlateau (patience=3)  | Reduces LR when validation plateaus           |

#### 3.2 Training Configuration

| Parameter     | Value                    |
|---------------|--------------------------|
| Image Size    | 224 x 224                |
| Epochs        | 10                       |
| Batch Size    | 8                        |
| Learning Rate | 0.001                    |
| Device        | CPU (AMD Ryzen 3 3250U)  |
| Workers       | 0 (Windows requirement)  |
| Training Time | ~198 minutes (3.3 hours) |

#### 3.3 Data Augmentation

Safe medical augmentations applied during training:

| Augmentation    | Value      | Status                                  |
|-----------------|------------|-----------------------------------------|
| Horizontal Flip | p=0.5      | Allowed — chest X-rays are symmetric    |
| Small Rotation  | ±5 degrees | Allowed — patient positioning variation |

| Augmentation        | Value             | Status                                  |
|---------------------|-------------------|-----------------------------------------|
| Mild Zoom           | 0.95 - 1.05 scale | Allowed — scanner distance variation    |
| Brightness/Contrast | ±10%              | Allowed — scanner calibration variation |
| Vertical Flip       | Disabled          | Not Allowed — anatomically incorrect    |
| Color Jitter        | Disabled          | Not Allowed — grayscale medical images  |

3.4 Results

| Metric                  | Score  | Assessment                                                       |
|-------------------------|--------|------------------------------------------------------------------|
| Validation AUC-ROC      | 0.8858 | Excellent — model ranks sick vs normal correctly 88% of the time |
| Validation F1-Score     | 0.1992 | Moderate — improving across epochs                               |
| Best Val F1 (saved)     | 0.3279 | Best checkpoint saved                                            |
| Training Loss (final)   | 0.1651 | Decreasing — model learning correctly                            |
| Validation Loss (final) | 0.1700 | Stable — no overfitting detected                                 |

3.5 Preprocessing Details

- Border crop: 2% removed from each edge on-the-fly to eliminate scanner artifacts
- Normalization: ImageNet mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]
- Grayscale to 3-channel conversion for EfficientNet compatibility
- Resize to 224x224 pixels

## 4. Detection Model

### 4.1 Architecture

| Component         | Detail                           | Reason                                                  |
|-------------------|----------------------------------|---------------------------------------------------------|
| Model             | YOLOv8s (Small)                  | Document recommendation — balance of speed and accuracy |
| Pretrained On     | COCO dataset                     | Transfer learning for object detection                  |
| Classes           | 14 chest abnormalities           | VinBigData class set                                    |
| Annotation Format | YOLO format (.txt)               | Normalized center_x, center_y, width, height            |
| Loss Functions    | Box loss + Class loss + DFL loss | YOLOv8 built-in multi-component loss                    |
| Optimizer         | Adam                             | Adaptive learning rate optimization                     |

### 4.2 Folder Structure (Mandatory — Document §3)

The following folder structure was implemented as required by the project guidelines:

| Path                  | Contents                                 |
|-----------------------|------------------------------------------|
| dataset/images/train/ | 3,500 training PNG images                |
| dataset/images/val/   | 750 validation PNG images                |
| dataset/images/test/  | 750 test PNG images                      |
| dataset/labels/train/ | YOLO .txt label files for training set   |
| dataset/labels/val/   | YOLO .txt label files for validation set |
| dataset/labels/test/  | YOLO .txt label files for test set       |
| dataset/dataset.yaml  | YOLOv8 dataset configuration file        |

### 4.3 YOLO Annotation Format

Each .txt label file contains one line per bounding box in the following format:

```
class_id center_x center_y width height
```

All values are normalized between 0 and 1 relative to image dimensions. Labels were generated from train.csv bounding box coordinates and scaled from original DICOM resolution to 224x224.

## 4.4 Training Configuration

| Parameter                 | Value                    | Document Reference                   |
|---------------------------|--------------------------|--------------------------------------|
| Image Size                | 224 x 224                | Adjusted for CPU hardware (doc: 640) |
| Epochs                    | 40 (30 + 10 fine-tuning) | Doc recommends 50-80                 |
| Batch Size                | 8                        | Reduced for 8GB RAM constraint       |
| Learning Rate (initial)   | 0.001                    | Document §8                          |
| Learning Rate (fine-tune) | 0.0001                   | Lower LR for fine-tuning phase       |
| IoU Threshold             | 0.5                      | Document §7: mAP@50                  |
| Confidence Threshold      | 0.25                     | Document §8                          |
| Device                    | CPU (AMD Ryzen 3 3250U)  | No CUDA support on hardware          |
| Workers                   | 0                        | Windows multiprocessing requirement  |
| Training Time             | ~323 minutes (5.4 hours) | Measured                             |

## 4.5 Data Augmentation (Medical Safe — Document §5)

| Augmentation          | Value       | Allowed? | Reason                                |
|-----------------------|-------------|----------|---------------------------------------|
| Horizontal Flip       | p=0.5       | Yes      | Chest X-rays are left-right symmetric |
| Rotation              | ±10 degrees | Yes      | Patient positioning variation         |
| Random Scaling        | 10%         | Yes      | Scanner distance variation            |
| Brightness Adjustment | ±10%        | Yes      | Scanner calibration variation         |
| Vertical Flip         | Disabled    | No       | Anatomically incorrect                |
| Perspective Warp      | Disabled    | No       | Distorts anatomy                      |
| Mosaic                | Disabled    | No       | Not medically realistic               |
| Mixup                 | Disabled    | No       | Not medically realistic               |
| Random Erasing        | Disabled    | No       | Document §5 explicitly not allowed    |
| Color Transformation  | Disabled    | No       | Grayscale medical images              |

## 4.6 Speed Optimizations (CPU)

- rect=True: Rectangular training — skips wasted padding computation (~15% faster)
- cos\_lr=True: Cosine learning rate scheduling — more efficient training per epoch
- amp=True: Mixed precision arithmetic — faster math, saves RAM
- torch.set\_num\_threads(4): Forces all 4 CPU cores for parallel computation
- cache=False: Disabled — only 0.9GB free RAM, caching would cause crash

4.7 Results

| Metric    | Score  | Assessment                                                       |
|-----------|--------|------------------------------------------------------------------|
| mAP@50    | 0.0658 | Low — primarily due to small image size (224 vs recommended 640) |
| mAP@50-95 | 0.0286 | Low — stricter COCO metric                                       |
| Precision | 0.3793 | Acceptable — 38% of predictions are correct                      |
| Recall    | 0.0821 | Low — model conservative in predictions                          |

4.8 Validation Checklist

| Check                        | Status | Notes                                    |
|------------------------------|--------|------------------------------------------|
| mAP improved from baseline   | Pass   | Model learned from random initialization |
| No excessive false positives | Pass   | Precision = 0.38                         |
| Bounding boxes generated     | Pass   | Boxes detected on test images            |
| Confidence scores reasonable | Pass   | Scores in expected range 0-1             |
| Logs and metrics saved       | Pass   | Auto-saved by YOLOv8 to yolo_output/     |
| PR Curve generated           | Pass   | Saved in training results folder         |
| Confusion Matrix generated   | Pass   | Saved in training results folder         |

## 5. Deliverables

### 5.1 Classification Model Deliverables

| File                       | Location                     | Description                        |
|----------------------------|------------------------------|------------------------------------|
| best_model.pt              | output/best_model.pt         | Best model weights (Val F1=0.3279) |
| training_curves.png        | output/training_curves.png   | Loss, AUC-ROC, F1 training curves  |
| data_verification.png      | output/data_verification.png | 10-sample data verification grid   |
| Classification_Model.ipynb | src/                         | Complete training notebook         |

### 5.2 Detection Model Deliverables

| File                         | Location                       | Description                          |
|------------------------------|--------------------------------|--------------------------------------|
| best.pt                      | yolo_output /best.pt           | Best YOLOv8s weights                 |
| results.png                  | yolo_output/yolov8s_cliniScan/ | Training losses + mAP curves         |
| confusion_matrix.png         | yolo_output/yolov8s_cliniScan/ | Per-class confusion matrix           |
| PR_curve.png                 | yolo_output/yolov8s_cliniScan/ | Precision-Recall curve               |
| F1_curve.png                 | yolo_output/yolov8s_cliniScan/ | F1 score curve                       |
| test_predictions_20.png      | yolo_output/                   | 20 test image predictions with boxes |
| bbox_verification.png        | yolo_output/                   | Bounding box data verification       |
| detection_summary_report.txt | yolo_output/                   | Auto-generated summary report        |
| Detection_Model.ipynb        | src/                           | Complete detection training notebook |

## 6. Challenges and Solutions

| Challenge                           | Impact                              | Solution Applied                                                               |
|-------------------------------------|-------------------------------------|--------------------------------------------------------------------------------|
| No GPU — CPU only training          | Very slow training speed            | Used EfficientNet-B0 + YOLOv8n/s, reduced batch size, used amp=True, rect=True |
| Only 8GB RAM                        | Training crashes if too much loaded | Disabled cache, batch=8, workers=0                                             |
| Only 0.9GB free RAM during training | Could not cache dataset             | cache=False, closed all background apps                                        |

| Challenge                                           | Impact                                           | Solution Applied                                                               |
|-----------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------|
| C: drive ran out of space during detection training | Training crashed at epoch 16                     | Freed 20GB by deleting dataset folder and temp files, resumed with last.pt     |
| imgsz=640 too slow on CPU                           | ~3 hours per epoch (6+ days total)               | Reduced to imgsz=224 (matches processed images), brought down to ~25 min/epoch |
| Class imbalance in dataset                          | Some rare conditions not detected                | Weighted sampling considered, BCEWithLogitsLoss handles multi-label            |
| PyTorch set_num_interop_threads error               | Could not set CPU threads after training started | Removed the line, used set_num_threads only                                    |
| YOLO resume lost dataset path                       | FileNotFoundError on resume                      | Rebuilt dataset folder before resuming                                         |

## 7. Model Comparison Summary

| Aspect        | Classification Model       | Detection Model        |
|---------------|----------------------------|------------------------|
| Architecture  | EfficientNet-B0            | YOLOv8s                |
| Task          | Multi-label classification | Bounding box detection |
| Image Size    | 224 x 224                  | 224 x 224              |
| Epochs        | 10                         | 40 (30 + 10 fine-tune) |
| Training Time | ~198 min (3.3 hrs)         | ~323 min (5.4 hrs)     |
| Key Metric    | AUC-ROC: 0.8858            | mAP@50: 0.0658         |
| Precision     | N/A (multi-label)          | 0.3793                 |
| Recall        | N/A (multi-label)          | 0.0821                 |
| F1 Score      | 0.3279 (best)              | 0.0740                 |
| Status        | Completed                  | Completed (40 epochs)  |

## 8. Conclusion

Both models were successfully trained on CPU hardware under significant memory and storage constraints. The Classification Model achieved a strong AUC-ROC of 0.8858, demonstrating effective learning of chest X-ray abnormality patterns. The Detection Model achieved a Precision of 0.3793, indicating that when it does detect a finding, it is correct approximately 38% of the time.

The lower mAP@50 score in detection (0.0658) is primarily attributed to the hardware constraint requiring imgsz=224 instead of the recommended 640 — smaller images make bounding boxes

proportionally much smaller and harder to localize accurately. This is a known limitation of training detection models on CPU with limited RAM.

All document guidelines were followed for both models including mandatory folder structure, YOLO annotation format, safe medical augmentations, and complete deliverables package.